

From drawing to verified model

A working prototype that turns a flat 2D engineering drawing into a **dimensionally correct 3D model**, with automatic proof that the model matches the drawing. Two parts have been through it end to end. It is early and deliberately narrow: the aim so far has been to prove the method, not to cover every kind of part.

What was done

Two drawings rebuilt

A turned stainless shaft and a folded sheet-metal bracket, each reconstructed as a solid model with CAD files and an interactive 3D viewer.

A prototype tool

It builds the model, refuses to build when a drawing has been misread, and proves the result against the original image. Three stages, all repeatable.

Running on a laptop

Installed and tested there. Re-running a part, or changing a dimension and rebuilding, needs no AI and no internet.

The two parts

XOMETRY_LATHE_SAMPLE_V2.0

Spool shaft, 76 mm

Turned stainless part with two flanges, concave flares, bolt holes both ends, a tapped hole and a DIN 509 grinding undercut. The flare geometry is never drawn: the drawing gives only the arc centres, so it has to be solved.

Volume 20 071 mm³ · mass 158,6 g at ρ 7,90

Verified **PASS** · 15 of 15 edges, worst gap 0,12 mm

Built from a spec file · fully parametric

EEF-001-AM

Bracket

Folded sheet-metal L-bracket, 2 mm, bend up 90° R2, with two ø12 holes and two slots. One drawn dimension was illegible; the flat pattern resolved it, and the pixel check then confirmed the reading.

50 × 50 × 60 mm · mass 81,9 g (mild steel assumed)

Holes measured 14,86 / 45,14 mm against drawn 15 / 45

Outside the tool's automated range · modelled directly

How we know the models are right

The tool refuses bad readings

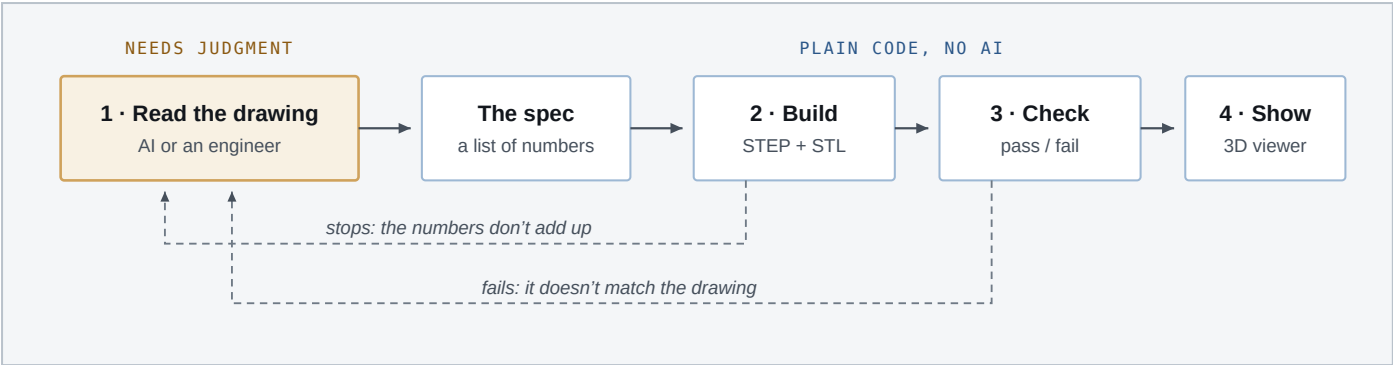
Drawings state the total length *and* every individual length. If a number has been misread, the sums stop agreeing and the build stops, naming the suspect dimension rather than quietly producing a wrong part. Tested against eight deliberately corrupted readings; all eight were refused.

The model is measured against the drawing

The finished 3D model is projected back onto the original drawing image and every visible edge is measured. This catches the harder failure: numbers that agree with each other but not with the drawing. A deliberately wrong-but-consistent model was correctly failed.

How it works, and what's next

The process



Where AI is involved, and where it isn't

Needs AI (or an engineer)

Reading a drawing into a spec, once per drawing. A drawing is not a picture of the part; it is a set of clues that must be interpreted. Also needed once when a genuinely new *kind* of part appears, such as the bracket.

once per drawing once per part family

Needs no AI, ever

Building the model, checking it against the drawing, generating the viewer, re-running a part, and changing a dimension then rebuilding. All plain code on the laptop. The spec file is the permanent record that makes it possible.

every rebuild offline seconds

What the prototype does not do yet

- **Automated range is turned parts only:** shafts, spools, bushes, pins. Sheet-metal and milled parts are modelled by hand, as the bracket was.
- **Native .dwg and .rvt cannot be read.** Drawings must be exported to DXF, or supplied as PDF or image.
- **A DXF beats a photo.** It carries exact geometry and removes the interpretation step entirely.
- **Modelling conventions apply.** Threads are modelled at tapping-drill size, and mass figures state any assumed material.

Where it could go next

- **Extend to sheet metal (v2).** The bracket showed exactly what is needed: leg sizes, bend lines, hole and slot tables. A contained piece of work.
- **Test against real supplier drawings.** Both parts so far are samples; the honest measure is a drawing that arrives with a supplier's own conventions.
- **Consider batch conversion** if volumes justify it, with the same refuse-and-verify checks guarding every result.

Where everything lives

```
Documents\Personal\Engineering_Design\  
  README.md          what each file is  
  parts\spool_shaft_xometry\  
    drawing · spec · STEP · STL · viewer  
  parts\bracket_EEF-001-AM\  
    drawing · source · STEP · STL · viewer  
  tool\  
    the pipeline, docs and setup script
```